A SCHEME COLIMIT IS ALSO THE COLIMIT OF ALGEBRAIC SPACES UNDER FOUR HYPOTHESES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 333-347 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 349-376, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 16 of queue_spaces-pushouts.json (pb=1666, stmt 333-347, proof 349-376).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{I} \to (Sch/S)_{fppf}$, $i \mapsto X_i$ be a diagram of schemes over S as above. Assume that*

(1) *$X = \operatorname{colim} X_i$ exists in the category of schemes,*
(2) *$\coprod X_i \to X$ is surjective,*
(3) *if $U \to X$ is étale and $U_i = X_i \times_X U$, then $U = \operatorname{colim} U_i$ in the category of schemes, and*
(4) *every object $(U_i \to X_i)$ of $\lim X_{i,étale}$ with $U_i \to X_i$ separated is in the essential image of the functor $X_{étale} \to \lim X_{i,étale}$.*

Then $X = \operatorname{colim} X_i$ in the category of algebraic spaces over S also.

Proof. Let Z be an algebraic space over S. Suppose that $f_i : X_i \to Z$ is a family of morphisms such that for each $i \to j$ the composition $X_i \to X_j \to Z$ is equal to f_i. We have to construct a morphism of algebraic spaces $f : X \to Z$ such that we can recover f_i as the composition $X_i \to X \to Z$. Let $W \to Z$ be a surjective étale morphism of a scheme to Z. We may assume that W is a disjoint union of affines and in particular we may assume that $W \to Z$ is separated. For each i set $U_i = W \times_{Z,f_i} X_i$ and denote $h_i : U_i \to W$ the projection. Then $U_i \to X_i$ forms an object of $\lim X_{i,étale}$ with $U_i \to X_i$ separated. By assumption (4) we can find an étale morphism $U \to X$ and (functorial) isomorphisms $U_i = X_i \times_X U$. By assumption (3) there exists a morphism $h : U \to W$ such that the compositions $U_i \to U \to W$ are h_i. Let $g : U \to Z$ be the composition of h with the map $W \to Z$. To finish the proof we have to show that $g : U \to Z$ descends to a morphism $X \to Z$.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

To do this, consider the morphism $(h, h) : U \times_X U \to W \times_S W$. Composing with $U_i \times_{X_i} U_i \to U \times_X U$ we obtain (h_i, h_i) which factors through $W \times_Z W$. Since $U \times_X U$ is the colimit of the schemes $U_i \times_{X_i} U_i$ by (3) we see that (h, h) factors through $W \times_Z W$. Hence the two compositions $U \times_X U \to U \to W \to Z$ are equal. Because each $U_i \to X_i$ is surjective and assumption (2) we see that $U \to X$ is surjective. As Z is a sheaf for the étale topology, we conclude that $g : U \to Z$ descends to $f : X \to Z$ as desired. $\square$